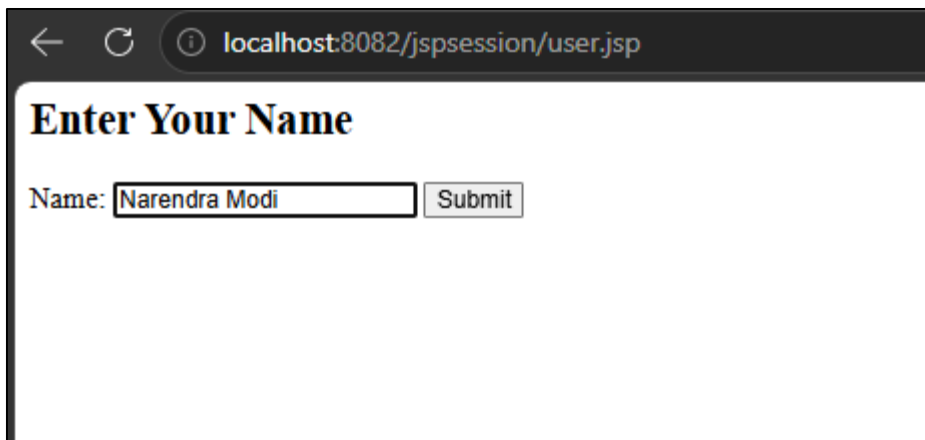


Assignment No	: 6
Roll No	: 381
Batch	: E4
PRN No	: 202502040032
Title of the Assignment	: JSP

Question 1: a) Write a program based on [JSP](#) actions and Objects

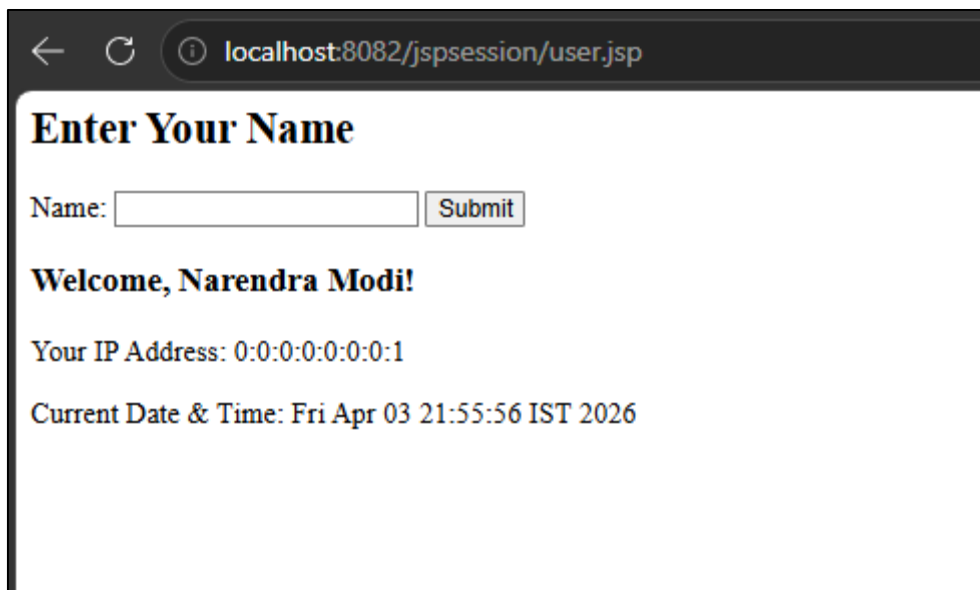
<i>user.jsp</i> <pre> <%@ page import="java.util.*" %> <html> <head> <title>JSP Actions and Objects Demo</title> </head> <body> <h2>Enter Your Name</h2> <form method="post"> Name: <input type="text" name="name" required> <input type="submit" value="Submit"> </form> <% String name = request.getParameter("name"); if(name != null) { %> <jsp:include page="response.jsp" /> <% } %> </body> </html> </pre>
<i>Response.jsp</i> <pre> <%@ page import="java.util.*" %> <% String name = request.getParameter("name"); %> <h3>Welcome, <%= name %>!</h3> <p>Your IP Address: <%= request.getRemoteAddr() %></p> <p>Current Date & Time: <%= new Date() %></p> </pre>
<i>Outputs Screenshots: captions (all servlet and index.html)</i>



← ↻ ⓘ localhost:8082/jspsession/user.jsp

Enter Your Name

Name:



← ↻ ⓘ localhost:8082/jspsession/user.jsp

Enter Your Name

Name:

Welcome, Narendra Modi!

Your IP Address: 0:0:0:0:0:0:1

Current Date & Time: Fri Apr 03 21:55:56 IST 2026

Question 2: b) Implement a program for cookies handling and session tracking.

Index.jsp

```
<%@ page import="java.util.*" %>
<%@ page import="jakarta.servlet.http.*" %>
<html>
<head>
  <title>User Dashboard</title>

  <style>
    body {
      font-family: Arial;
      background: linear-gradient(to right, #4facfe, #00f2fe);
      margin: 0;
    }

    .container {
      width: 400px;
      margin: 80px auto;
    }
  </style>
</head>
<body>
  <div class="container">
    <h1>User Dashboard</h1>
  </div>
</body>
</html>
```

```
background: white;
padding: 25px;
border-radius: 10px;
text-align: center;
box-shadow: 0 0 15px rgba(0,0,0,0.2);
}

input[type="text"] {
padding: 8px;
width: 80%;
margin: 10px 0;
}

input[type="submit"] {
padding: 10px 20px;
background: #4facfe;
color: white;
border: none;
border-radius: 5px;
cursor: pointer;
}

h2 {
color: #333;
}

.info {
margin-top: 15px;
text-align: left;
}
</style>
</head>

<body>

<div class="container">

<h2>User Dashboard</h2>

<form method="post">
  Enter Your Name:<br>
  <input type="text" name="username" required><br>
  <input type="submit" value="Save">
</form>

<%
String username = request.getParameter("username");

Integer visitCount = (Integer) session.getAttribute("visitCount");

if (visitCount == null) {
  visitCount = 1;
} else {
  visitCount = visitCount + 1;
}

session.setAttribute("visitCount", visitCount);

Cookie userCookie = null;
```

```
Cookie cookies[] = request.getCookies();

if (cookies != null) {
    for (int i = 0; i < cookies.length; i++) {
        if (cookies[i].getName().equals("username")) {
            userCookie = cookies[i];
        }
    }
}

if (username != null) {
    userCookie = new Cookie("username", java.net.URLEncoder.encode(username, "UTF-8"));
    userCookie.setMaxAge(60*60*24);
    response.addCookie(userCookie);
}
%>

<div class="info">

<%
if (userCookie != null) {
%>
    <p><b>Welcome Back, <%= userCookie.getValue() %> </b></p>
<%
} else {
%>
    <p><b>Welcome, New User </b></p>
<%
}
%>

<p> Session Visits: <b><%= visitCount %></b></p>
<p> Session ID: <%= session.getId() %></p>
<p> Current Time: <%= new Date() %></p>

</div>

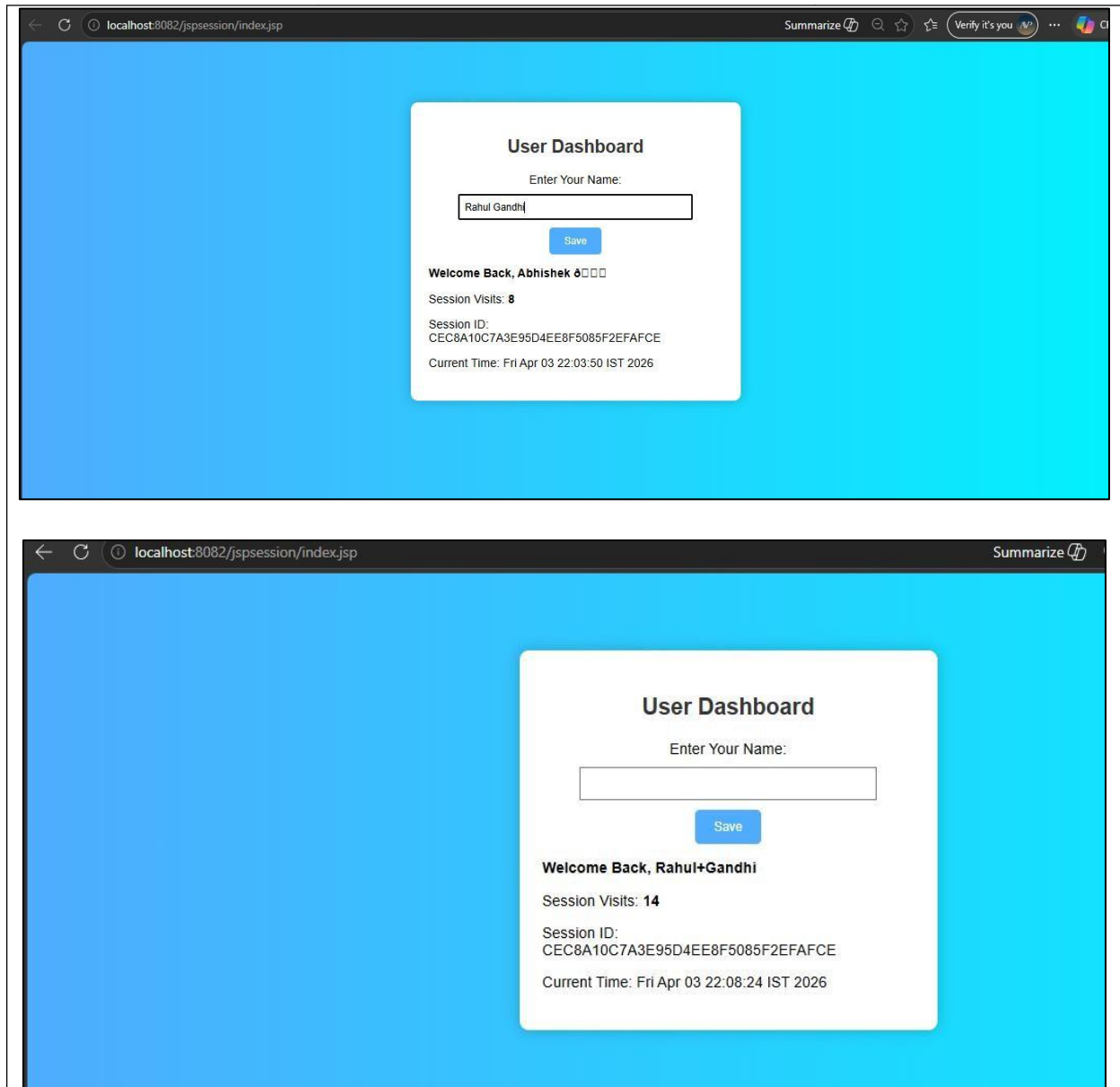
</div>

</body>
</html>
```

Logout.jsp

```
<%
    session.invalidate();
    response.sendRedirect("index.jsp");
%>
```

Outputs Screenshots: captions (all servlet and index.html)



Question 3: c) Write a program to demonstrate the core, format, function, and SQL tags of JSTL

Jstl.jsp

```
<%@ page contentType="text/html; charset=UTF-8"%>

<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c"%>
<%@ taglib uri="http://java.sun.com/jsp/jstl/fmt" prefix="fmt"%>
<%@ taglib uri="http://java.sun.com/jsp/jstl/functions" prefix="fn"%>
<%@ taglib uri="http://java.sun.com/jsp/jstl/sql" prefix="sql"%>

<html>
<head>
<title>JSTL Demo Application</title>
```

```
<style>
  body {
    font-family: Arial;
    background: linear-gradient(to right, #43cea2, #185a9d);
    color: white;
    text-align: center;
  }
  .box {
    background: white;
    color: black;
    padding: 20px;
    margin: 40px auto;
    width: 60%;
    border-radius: 10px;
  }
</style>
</head>

<body>

<div class="box">
  <h2> JSTL Core + Format + Function + SQL Demo</h2>

  <c:set var="name" value="Abhi Sharma" />
  <c:if test="${not empty name}">
    <p>Welcome, ${name}</p>
  </c:if>

  <c:forEach var="i" begin="1" end="3">
    <p>Iteration: ${i}</p>
  </c:forEach>

  <hr>

  <p><u>Uppercase</u> Name: ${fn:toUpperCase(name)}</p>
  <p>Length of Name: ${fn:length(name)}</p>

  <hr>

  <jsp:useBean id="now" class="java.util.Date" />
  <p>Current Date:
    <fmt:formatDate value="${now}" pattern="dd-MM-yyyy HH:mm:ss"/>
  </p>

  <fmt:setLocale value="en_IN"/>
  <fmt:formatNumber value="123456.78" type="currency"/>

  <hr>

  <sql:setDataSource var="db"
    driver="com.mysql.cj.jdbc.Driver"
    url="jdbc:mysql://localhost:3306/test"
    user="root"
    password="0744"
  />

  <sql:query dataSource="${db}" var="result">
    SELECT * FROM users;
  </sql:query>
```

```
<h3>Database Records:</h3>
```

```
<c:forEach var="row" items="${result.rows}">
```

```
<p>${row.id} - ${row.name}</p>
```

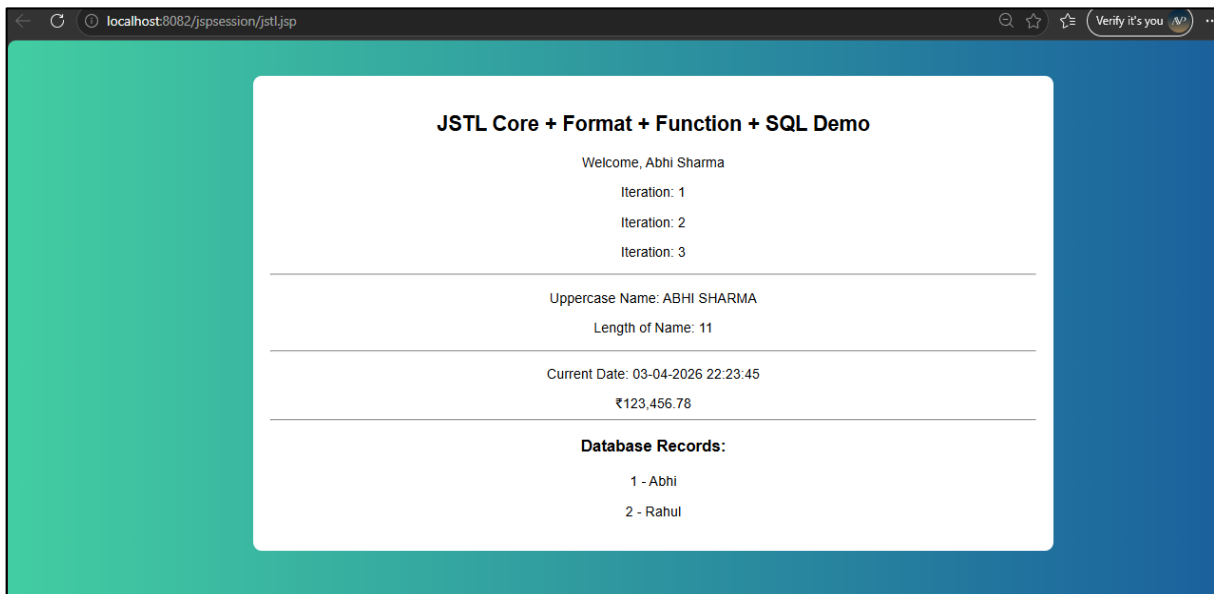
```
</c:forEach>
```

```
</div>
```

```
</body>
```

```
</html>
```

Outputs Screenshots: captions (all servlet and index.html)



Practice Problem:

1) Write a [JSP](#) program to create a simple user login page and display appropriate messages to demonstrate URLRewriting using sendRedirect .

Login.jsp

```
<html>
<head>
  <title>Login Page</title>

  <style>
    body {
      font-family: Arial;
      background: linear-gradient(to right, #667eea, #764ba2);
    }

    .box {
      width: 300px;
      margin: 100px auto;
      background: white;
      padding: 20px;
      border-radius: 10px;
      text-align: center;
    }

    input {
      margin: 10px;
      padding: 8px;
      width: 90%;
    }

    input[type="submit"] {
      background: #667eea;
      color: white;
      border: none;
    }
  </style>
</head>

<body>

<div class="box">
  <h2>Login</h2>

  <form action="process.jsp" method="post">
    <input type="text" name="username" placeholder="Enter Username" required><br>
    <input type="password" name="password" placeholder="Enter Password" required><br>
    <input type="submit" value="Login">
  </form>
</div>

</body>
</html>
```

Process.jsp

```
<%
String username = request.getParameter("username");
```



```
String password = request.getParameter("password");

if(username.equals("admin") && password.equals("1234"))
{
    response.sendRedirect("welcome.jsp?user=" + username);
}
else
{
    response.sendRedirect("error.jsp?msg=Invalid Username or Password");
}
%>
```

Welcome.jsp

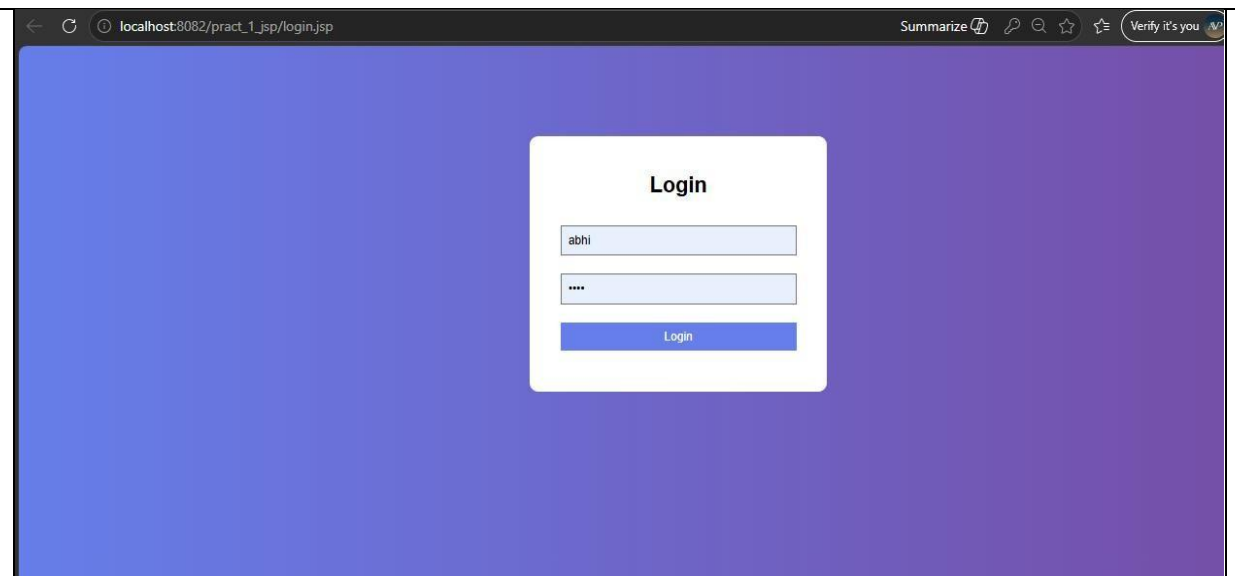
```
<html>
<body>

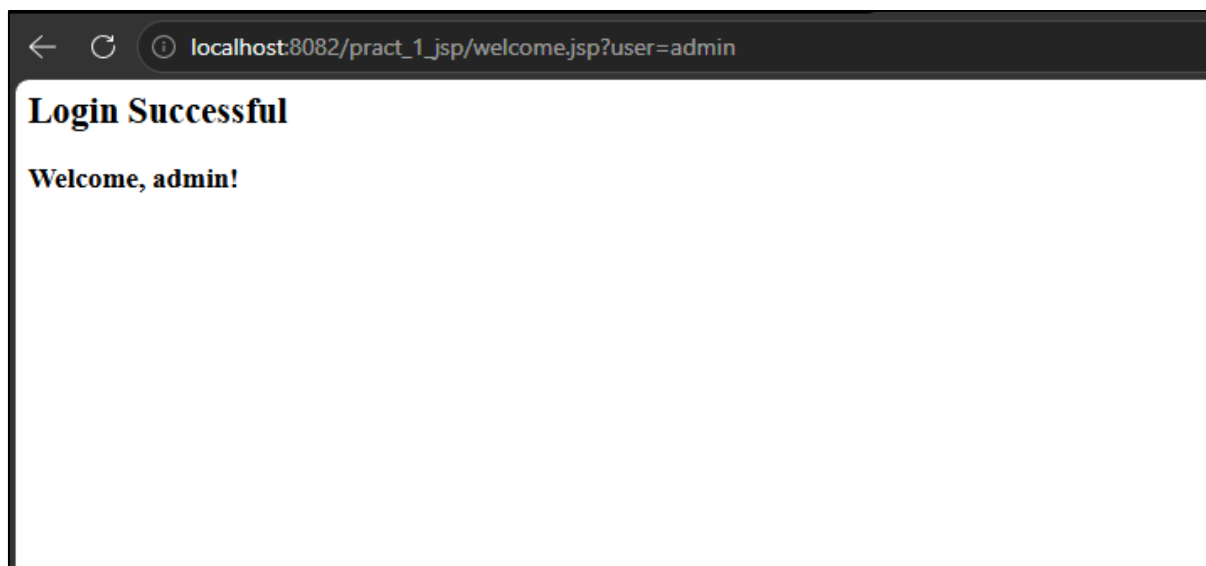
<%
String user = request.getParameter("user");
%>

<h2>Login Successful</h2>
<h3>Welcome, <%= user %>!</h3>

</body>
</html>
```

Outputs Screenshots: captions (all servlet and index.html)





2) Write a [JSP](#) program to calculate and pay the electricity bill generated given total units consumed and cost per unit as input from user

Electric_bill.jsp

```
<%@ page import="java.util.*" %>

<html>
<head>
  <title>Electricity Bill Generator</title>

  <style>
    body {
      font-family: Arial;
```

```
background: linear-gradient(to right, #ff9966, #ff5e62);
}

.container {
    width: 400px;
    margin: 80px auto;
    background: white;
    padding: 25px;
    border-radius: 10px;
    text-align: center;
    box-shadow: 0 0 15px rgba(0,0,0,0.2);
}

input {
    margin: 10px;
    padding: 8px;
    width: 90%;
}

input[type="submit"] {
    background: #ff5e62;
    color: white;
    border: none;
}

.result {
    margin-top: 15px;
    text-align: left;
}
</style>
</head>

<body>

<div class="container">

<h2> Electricity Bill Generator</h2>

<form method="post">
    Enter Units Consumed:<br>
    <input type="number" name="units" required><br>

    Cost Per Unit :<br>
    <input type="number" name="cost" step="0.01" required><br>

    <input type="submit" value="Generate Bill">
</form>

<%
String unitsStr = request.getParameter("units");
String costStr = request.getParameter("cost");

if(unitsStr != null && costStr != null)
{
    int units = Integer.parseInt(unitsStr);
    double cost = Double.parseDouble(costStr);

    double total = units * cost;

    // Optional: Add fixed charge (real-life touch)
    double fixedCharge = 50;
```

```
double finalBill = total + fixedCharge;
%>

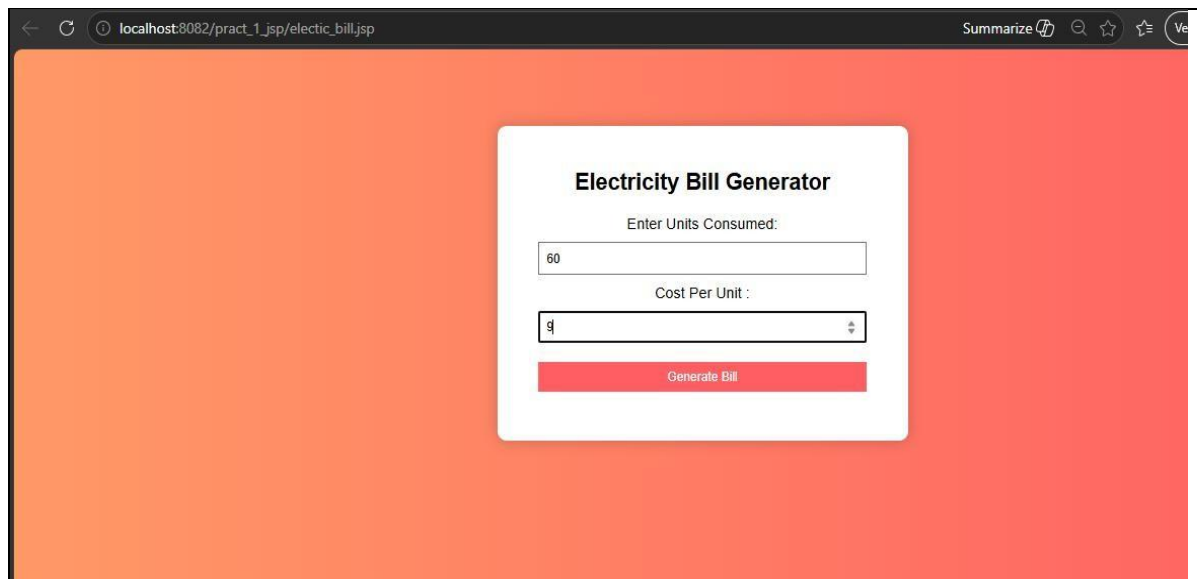
<div class="result">
  <h3>Bill Details</h3>
  <p>Units Consumed: <b><%= units %></b></p>
  <p>Cost per Unit: <b><%= cost %></b></p>
  <p>Energy Charge: <b><%= total %></b></p>
  <p>Fixed Charge: <b><%= fixedCharge %></b></p>
  <hr>
  <h3>Total Bill: <%= finalBill %></h3>
  <p>Date: <%= new Date() %></p>
</div>

<%
}
%>

</div>

</body>
</html>
```

Outputs Screenshots: captions (all servlet and index.html)



The screenshot shows a web browser window with the address bar displaying "localhost:8082/pract_1_jsp/electic_bill.jsp". The page has a solid orange background. In the center, there is a white rectangular form titled "Electricity Bill Generator". Inside the form, there are two input fields: "Enter Units Consumed:" with the value "60" and "Cost Per Unit :" with the value "9". Below these fields is a red button labeled "Generate Bill". The browser's top bar includes a "Summarize" button and several icons.

Electricity Bill Generator

Enter Units Consumed:

Cost Per Unit :

Generate Bill

Bill Details

Units Consumed: **60**

Cost per Unit: **9.0**

Energy Charge: **540.0**

Fixed Charge: **50.0**

Total Bill: 590.0

Date: Fri Apr 03 22:52:10 IST 2026

3) Write a [JSP](#) program to enter the marks of five subjects and display the result using the include directive.

Sub_marks.jsp

```
<%@ page import="java.util.*" %>
<html>
<head>
  <title>Student Result System</title>

  <style>
    body {
      font-family: Arial;
      background: linear-gradient(to right, #36d1dc, #5b86e5);
    }

    .box {
      width: 400px;
      margin: 80px auto;
      background: white;
      padding: 20px;
      border-radius: 10px;
      text-align: center;
    }
  </style>
</head>
<body>
```

```
}

input {
    margin: 8px;
    padding: 8px;
    width: 80%;
}

input[type="submit"] {
    background: #5b86e5;
    color: white;
    border: none;
}
</style>
</head>

<body>

<div class="box">
    <h2>Student Result Calculator</h2>

    <form method="post">
        <input type="number" name="m1" placeholder="Subject 1" required><br>
        <input type="number" name="m2" placeholder="Subject 2" required><br>
        <input type="number" name="m3" placeholder="Subject 3" required><br>
        <input type="number" name="m4" placeholder="Subject 4" required><br>
        <input type="number" name="m5" placeholder="Subject 5" required><br>

        <input type="submit" value="Calculate Result">
    </form>

    <%
String m1 = request.getParameter("m1");

if(m1 != null)
{
    %>

        <%@ include file="result.jsp" %>

    <%
    }
    %>

</div>

</body>
</html>
```

Result.jsp

```
<%
int s1 = Integer.parseInt(request.getParameter("m1"));
int s2 = Integer.parseInt(request.getParameter("m2"));
int s3 = Integer.parseInt(request.getParameter("m3"));
int s4 = Integer.parseInt(request.getParameter("m4"));
int s5 = Integer.parseInt(request.getParameter("m5"));

int total = s1 + s2 + s3 + s4 + s5;
double percentage = total / 5.0;
```

```
String result;
```

```
if (percentage >= 50)
```

```
    result = "Pass";
```

```
else
```

```
    result = "Fail";
```

```
%>
```

```
<hr>
```

```
<h3>Result</h3>
```

```
<p>Total Marks: <b><%= total %></b></p>
```

```
<p>Percentage: <b><%= percentage %></b></p>
```

```
<p>Status: <b><%= result %></b></p>
```

Outputs Screenshots: captions (all servlet and index.html)

Student Result Calculator

Student Result Calculator

Result

Total Marks: **442**

Percentage: **88.4%**

Status: **Pass**

Submitted by,
Uddhav Tour .